25901590 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCZIM201 - INDUSTRIAL MICROBIOLOGY AND FERMENTATION TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Precipitation helps in concentrating and ..... the product. [K] / [CO1]
2. A typical fermenter is usually consructed using ..... steel to avoid corrosion. [K] / [CO2]
3. In brewing, proteases help in ..... clarification. [K] / [CO4]
4. Using microbes for waste treatment is preferred beacuse they reduce environmental ..... [K] / [CO1]
5. Acetic acid bacteria grow best at ..... pH. [K] / [CO2]
6. Cellulase activity isaffected by ..... and ..... [K] / [CO4]
7. Selective media help isolate microbes with ..... characteristics. [K] / [CO1]
8. SSF generally produces more concentrated products due to low ..... volume. [K] / [CO2]
9. Ferric iron acts as an ..... agent in uranium bioleaching. [U] / [CO5]

10. The equipment used for submerged fermentation is called..... [K] / [CO1]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Define continous sterilization. [K] / [CO1]

12. Discuss the nutrient concentration in batch fermentations. [K] / [CO2]

13. State how are aerobic conditions essential for glutmic acid production. [K] / [CO4]

14. Explain the exponential phase preferred for inoculum transfer. [K] / [CO1]

15. Explain how does over fermentation lead to poor texture. [U] / [CO3]

16. Discuss how is MEOR applied in mature oil fields. [A] / [CO5]

17. Compare chemical and physical methods of immobilisation. [K] / [CO1]

18. Explain the creation circulation between riser and downcomer zones. [K] / [CO2]

19. Explain how is pH controlled during penicillin fermentation. [K] / [CO4]

20. Define rate of dilution in continous fermenter. [K] / [CO2]

21. Explain the causes of formation in fermentation. [K] / [CO1]

22. List the measures ensure consistant quality in oriental fermented products. [U] / [CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Discuss the limitations of copper bioleaching. [U] / [CO5]

24. Explain how traditional fermentation practices evolved into modern industrial fermentation techniques. [K] / [CO1]

25. Describe how to prevent contamination during cheese aging. [K] / [CO2]

26. Explain how does agitation affect morphology of streptomyces and antibiotic yield. [U] / [CO4]

27. Differentiate between positive and negative selection techniques. [K] / [CO1]

28. Compare ale and larger fermentation. [K] / [CO2]

29. Summarize the role of microbial selecion in gold bioleaching. [K] / [CO5]

[4x5 = 20]